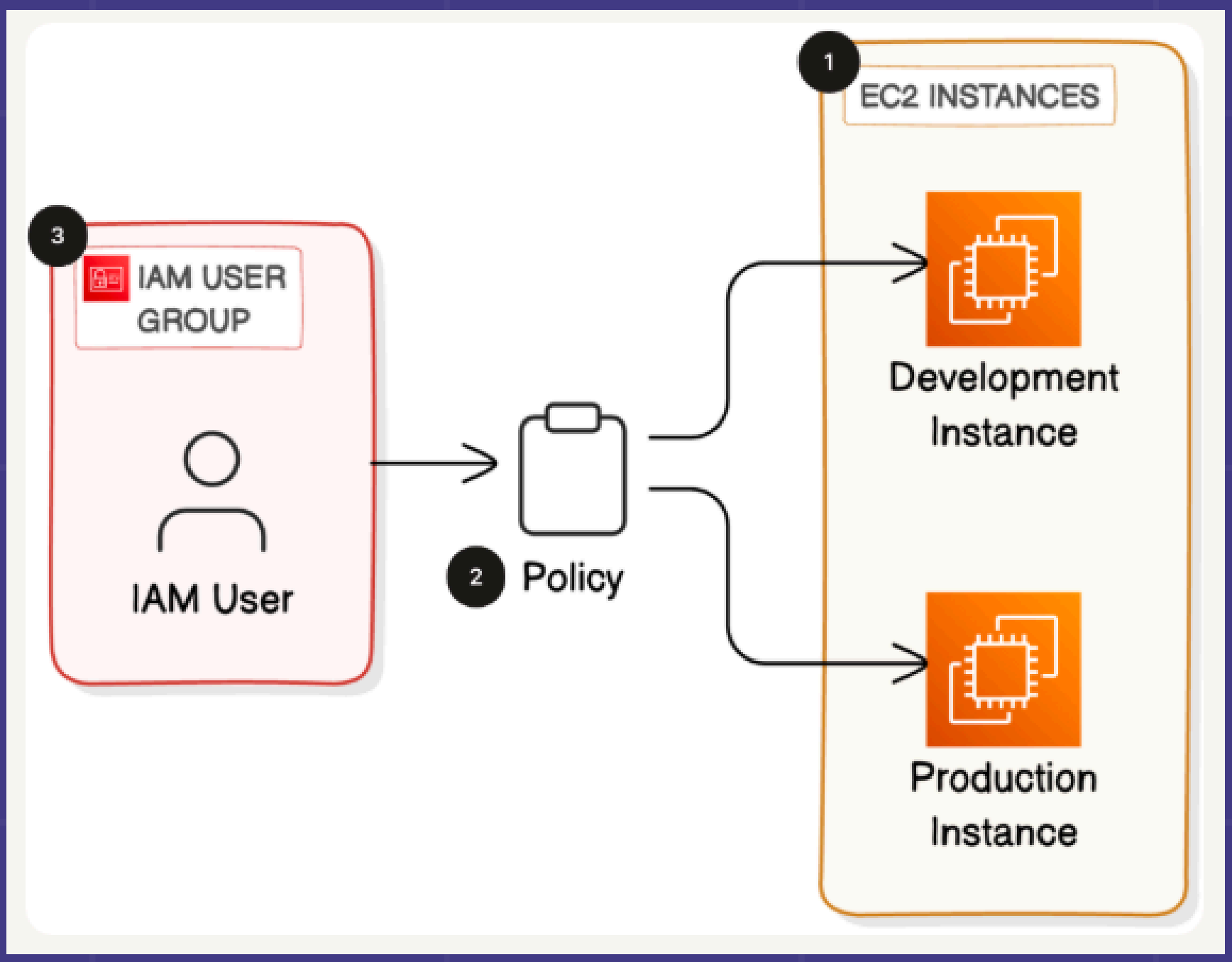
Cloud Security with

AWS IAM

Restricting Access to Production Using IAM Policies



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Project Overview

Project Goal

In this project, I built a simple cloud security control system using AWS IAM.

The goal was to:

- Allow developers to control Development servers
- Prevent them from accessing Production servers

This is a common real-world cloud security scenario used in companies.

Services Used

- AWS IAM
- Amazon EC2
- IAM Policies (JSON)

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Creating Development and Production Servers

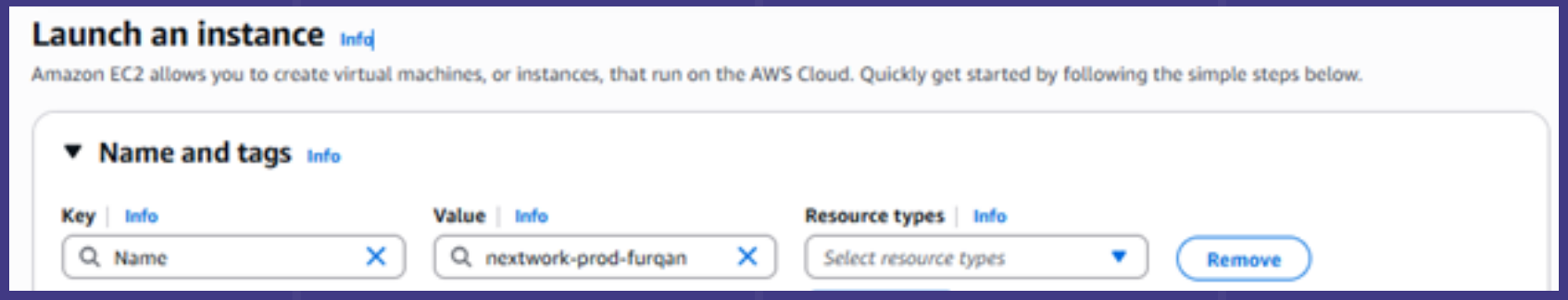
Launching EC2 Instances

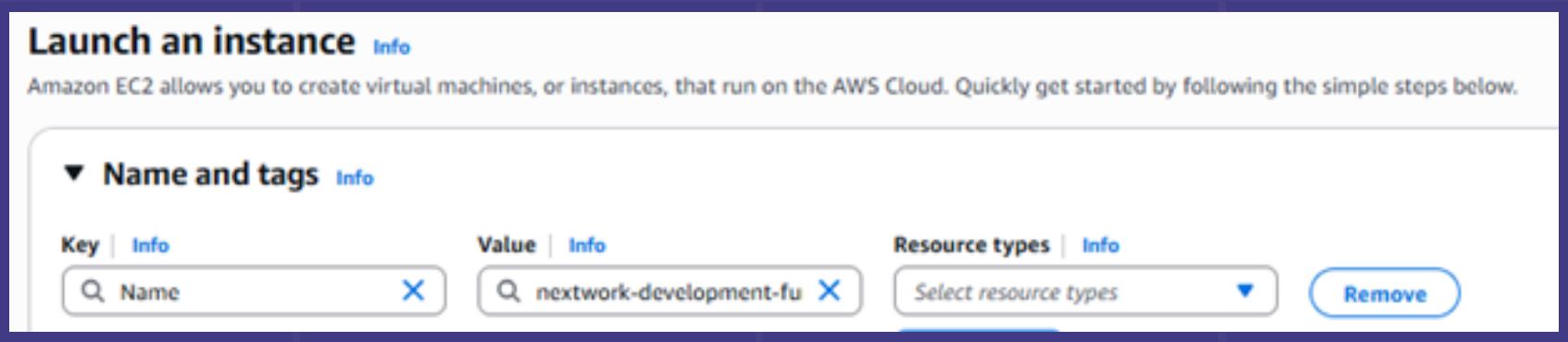
What is Amazon EC2?
Amazon EC2 (Elastic Compute Cloud) allows us to create virtual machines in the cloud.

These servers can run applications, websites, or backend systems.

For this project, I created:

- 1 Development EC2 Instance
- 1 Production EC2 Instance





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Organizing Cloud Resources with AWS Tags

What are AWS Tags?

AWS Tags are key–value labels you attach to AWS resources to organize, manage, and control them easily.

Example:
Key → Env
Value → Development

Key → Env
Value → Production

In this project, tags helped me separate development and production infrastructure.

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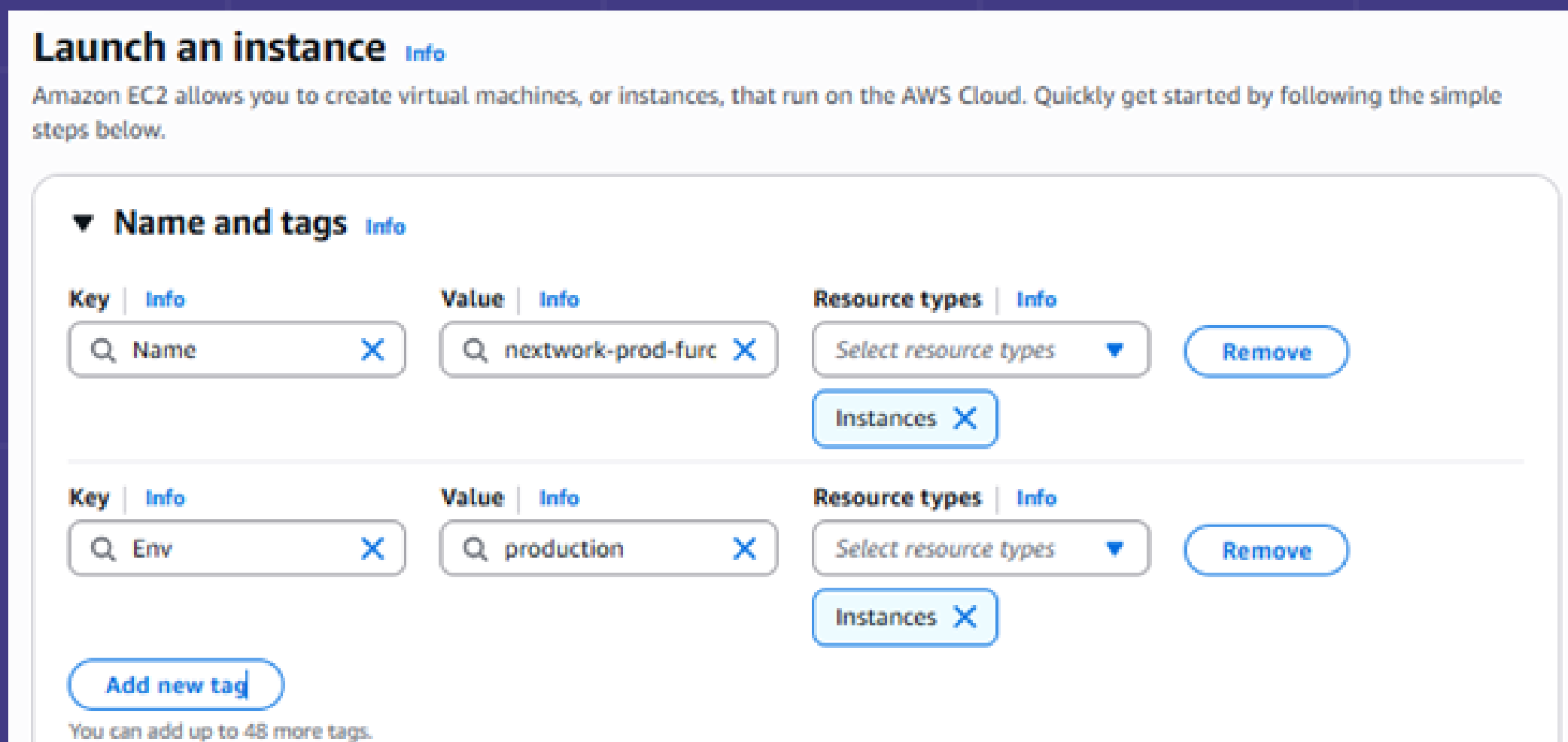
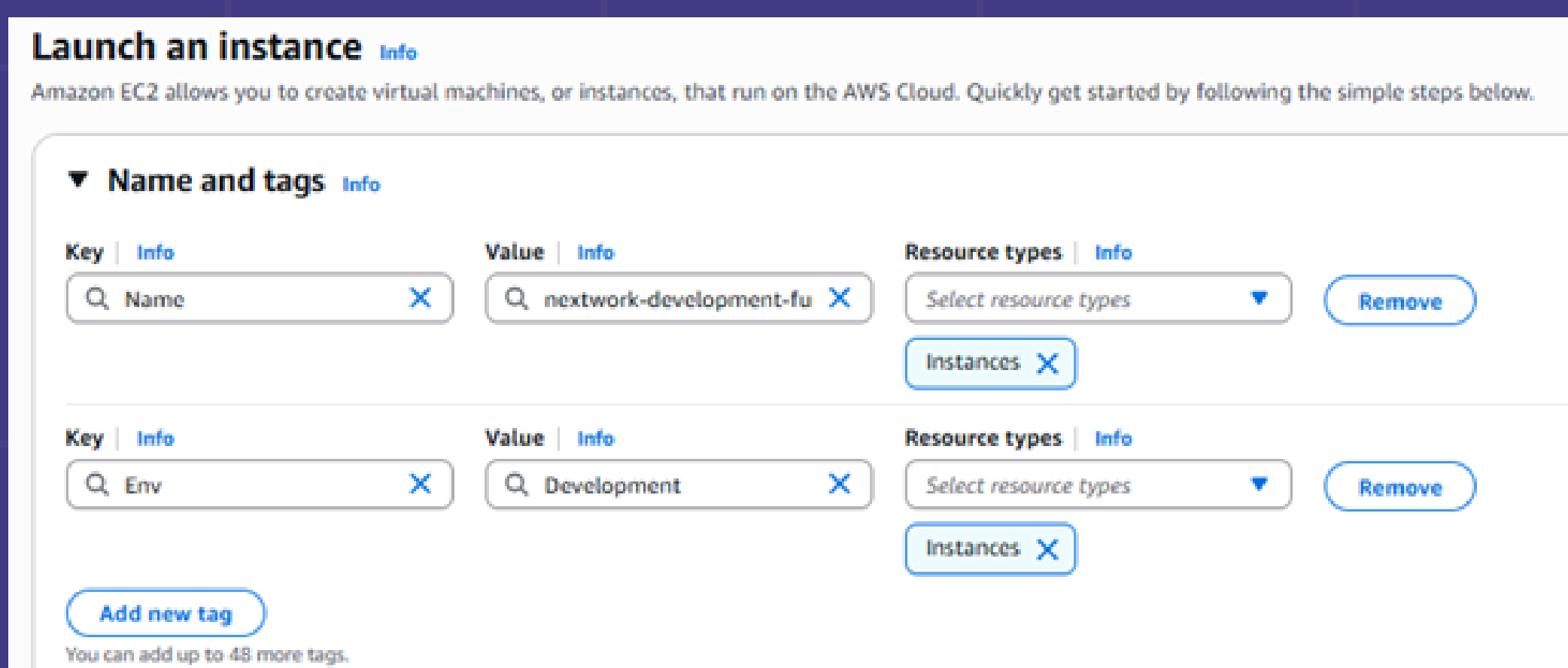
Organizing Cloud Resources with AWS Tags

Screenshots:





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IAM Policies

What are IAM Policies?

IAM Policies define permissions for AWS users or groups. They specify:

- Effect: Allow or Deny
- Action: The specific operation(s) to allow or deny
- Resource: The specific AWS resource(s) the policy applies to

For this project, I created a custom JSON policy:

- Allows all EC2 actions only on instances with the tag Env: Development
- Denies creating or deleting tags on all EC2 instances
- Allows describing EC2 instances on all resources

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IAM Policies

Screenshot:

Specify permissions Info

Add permissions by selecting services, actions, resources, and conditions. Build permission stater

Policy editor

```
1  {
2    "Version": "2012-10-17",
3    "Statement": [
4      {
5        "Effect": "Allow",
6        "Action": "ec2:*",
7        "Resource": "*",
8        "Condition": {
9          "StringEquals": {
10           "ec2:ResourceTag/Env": "Development"
11         }
12       }
13     },
14     {
15       "Effect": "Allow",
16       "Action": "ec2:Describe*",
17       "Resource": "*"
18     },
19     {
20       "Effect": "Deny",
21       "Action": [
22         "ec2:DeleteTags",
23         "ec2:CreateTags"
24       ],
25       "Resource": "*"
26     }
27   ]
28 }
29
```

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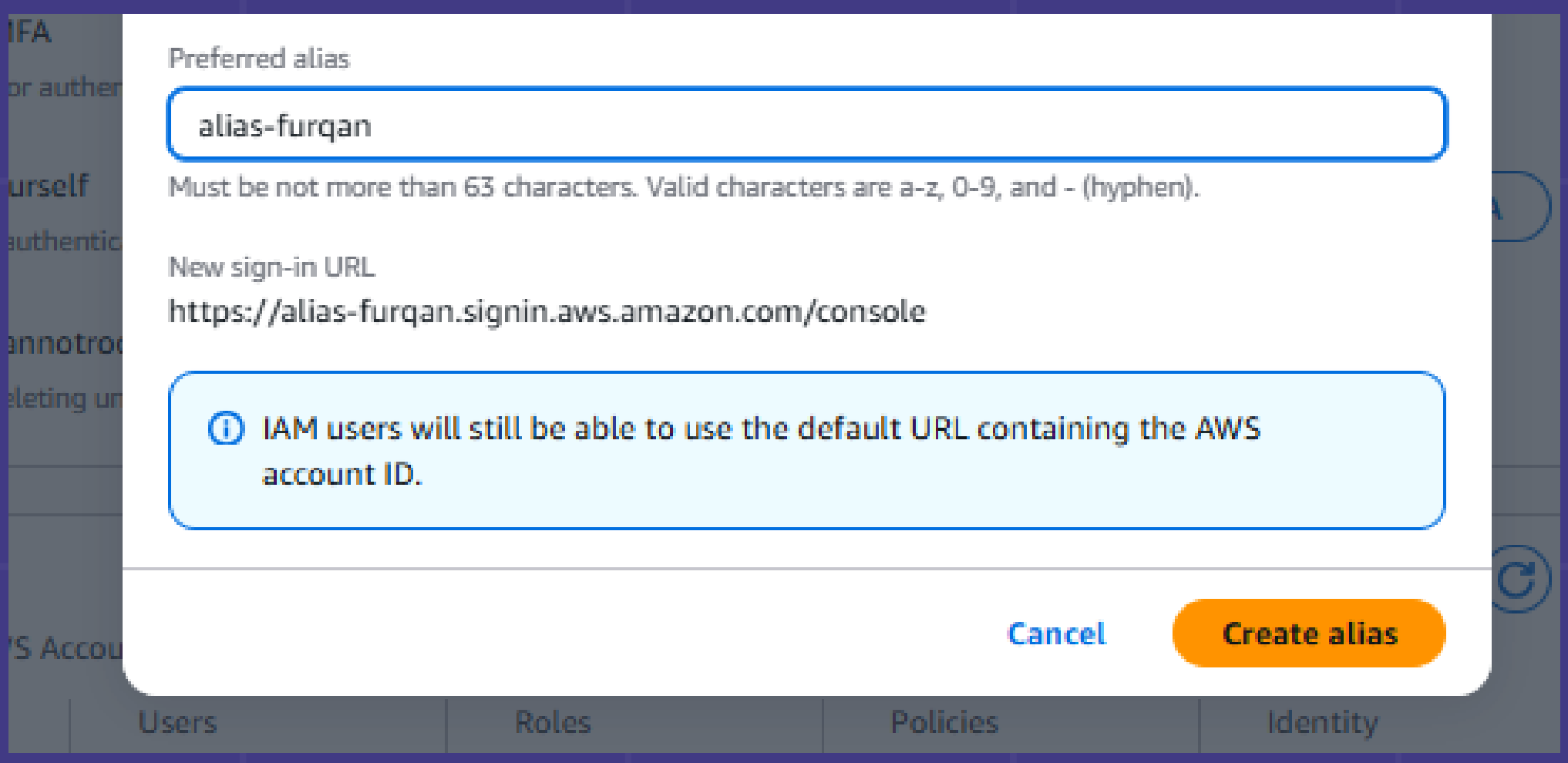
IAM Account Alias

What is an Account Alias?

An AWS Account Alias is a custom name assigned to your AWS account. It replaces the default account ID in the login URL, making sign-in easier for new users.

My Alias: alias-furqan

Screenshot:



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IAM Users & User Groups

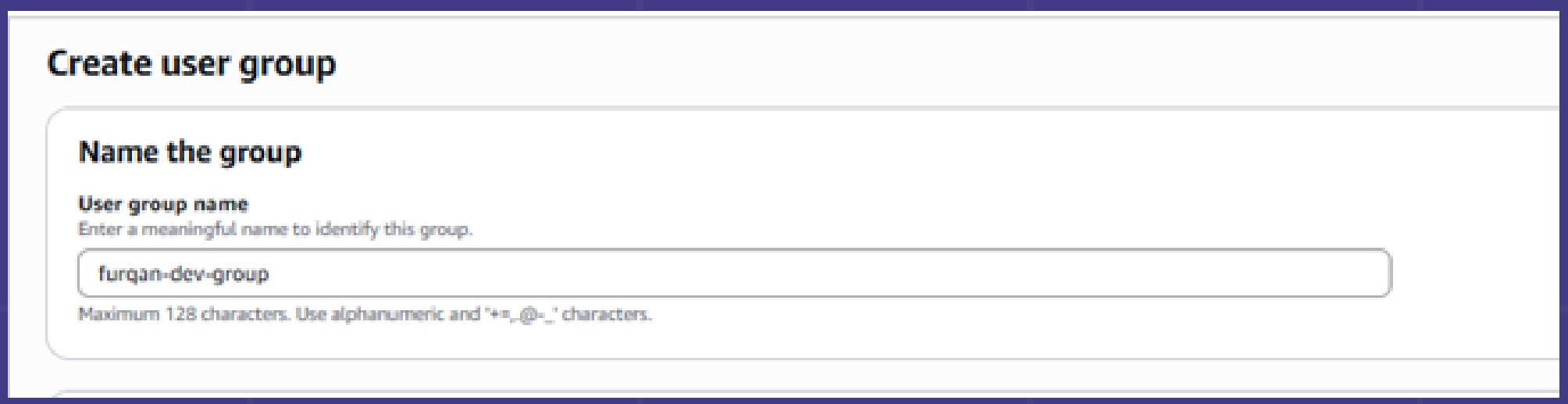
User Groups allow us to manage permissions for multiple users at once.

Instead of assigning permissions to each user individually, we attach policies to a group.

Any user added to that group automatically inherits those permissions.

For this project, I created a group called: furqan-dev-group

Screenshot:



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IAM Users & User Groups

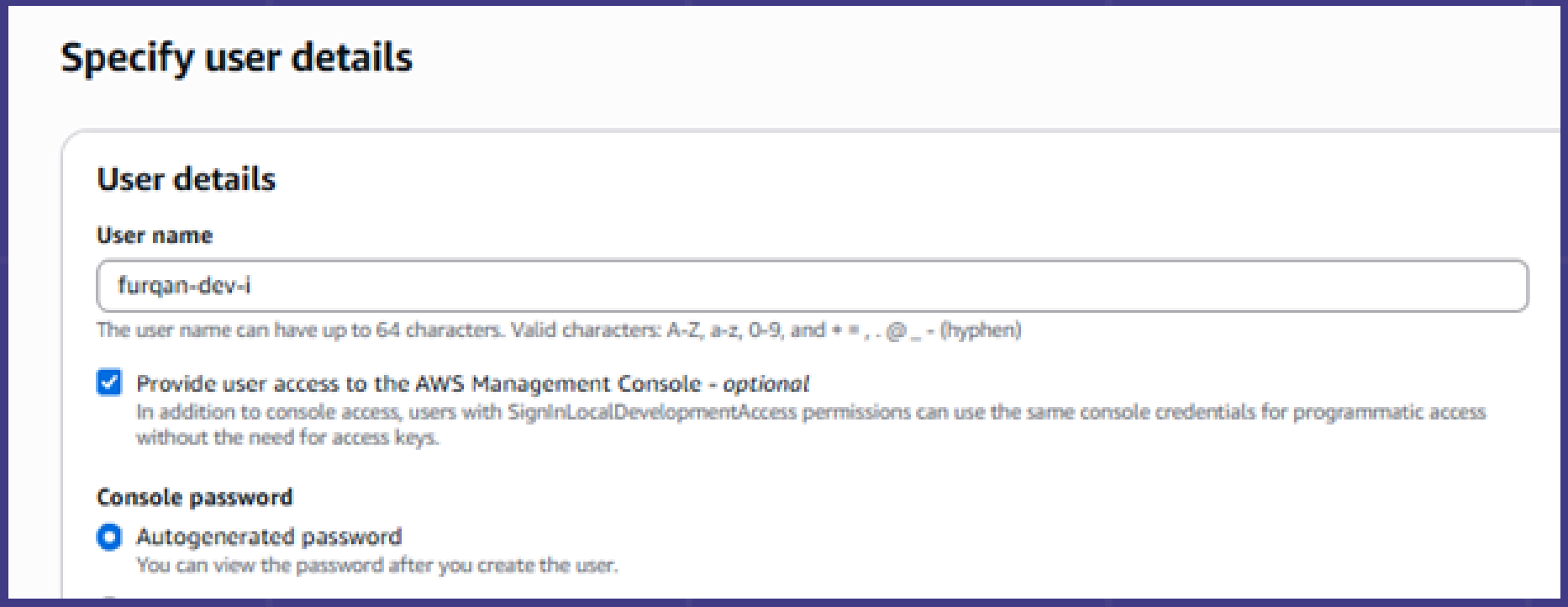
IAM Users are individual accounts created inside an AWS account.

These accounts allow people to log in and access AWS resources.

Each user has:
- Username
- Login credentials
- Assigned permissions

For this project, I created a user called: furqan-dev-i

Screenshot:



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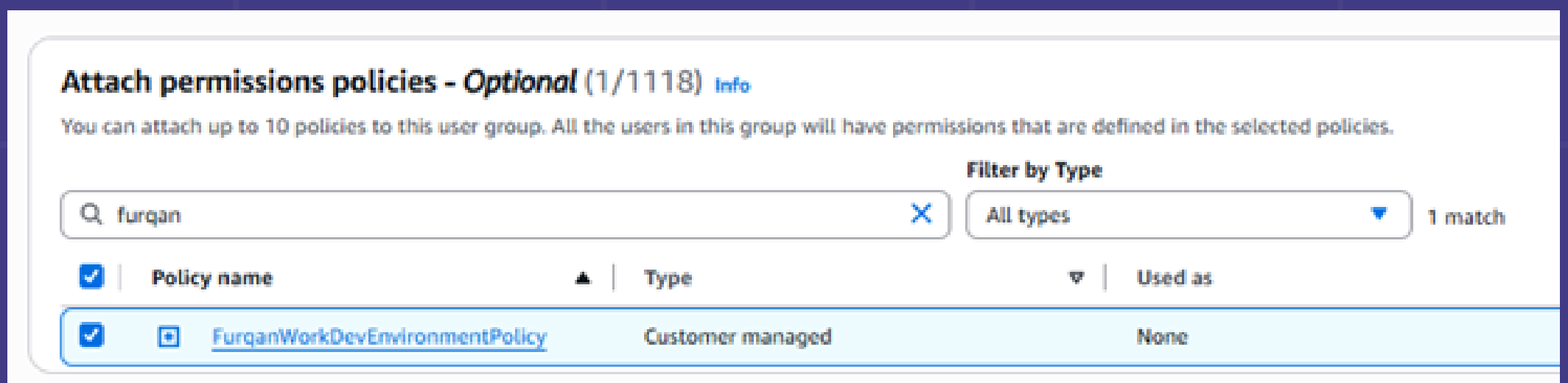
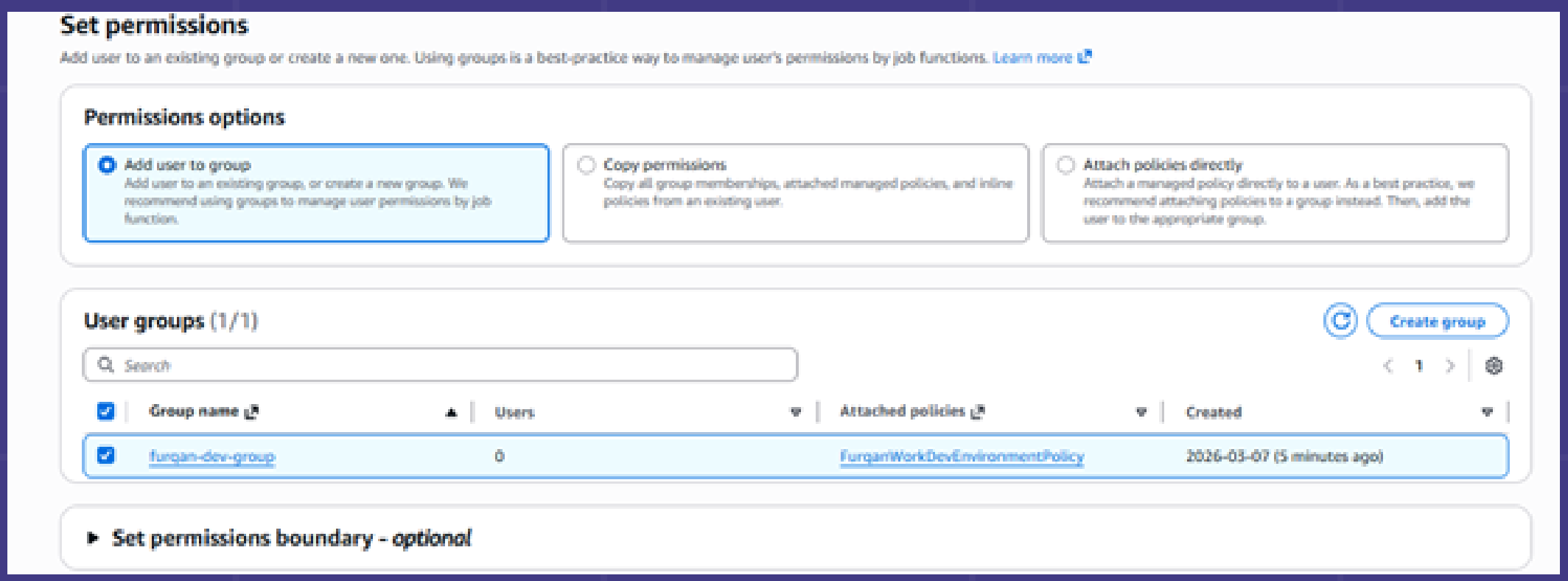
IAM Users & User Groups

Screenshots:





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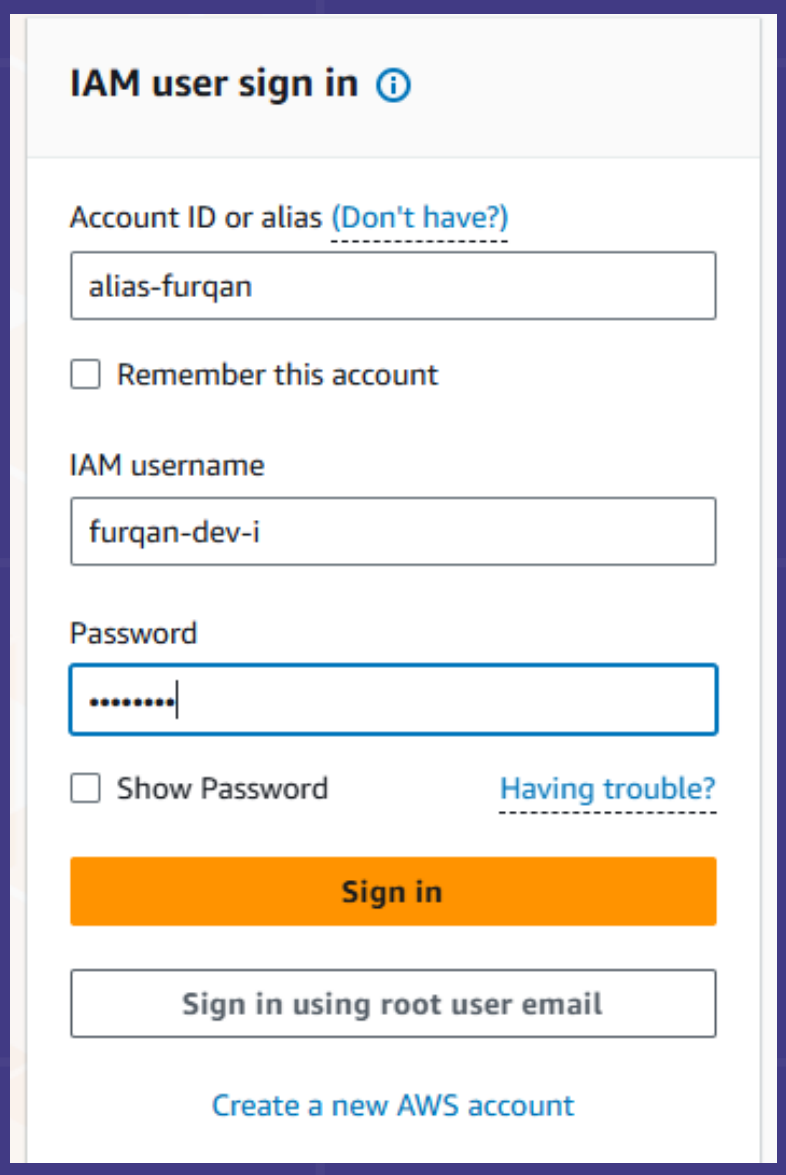
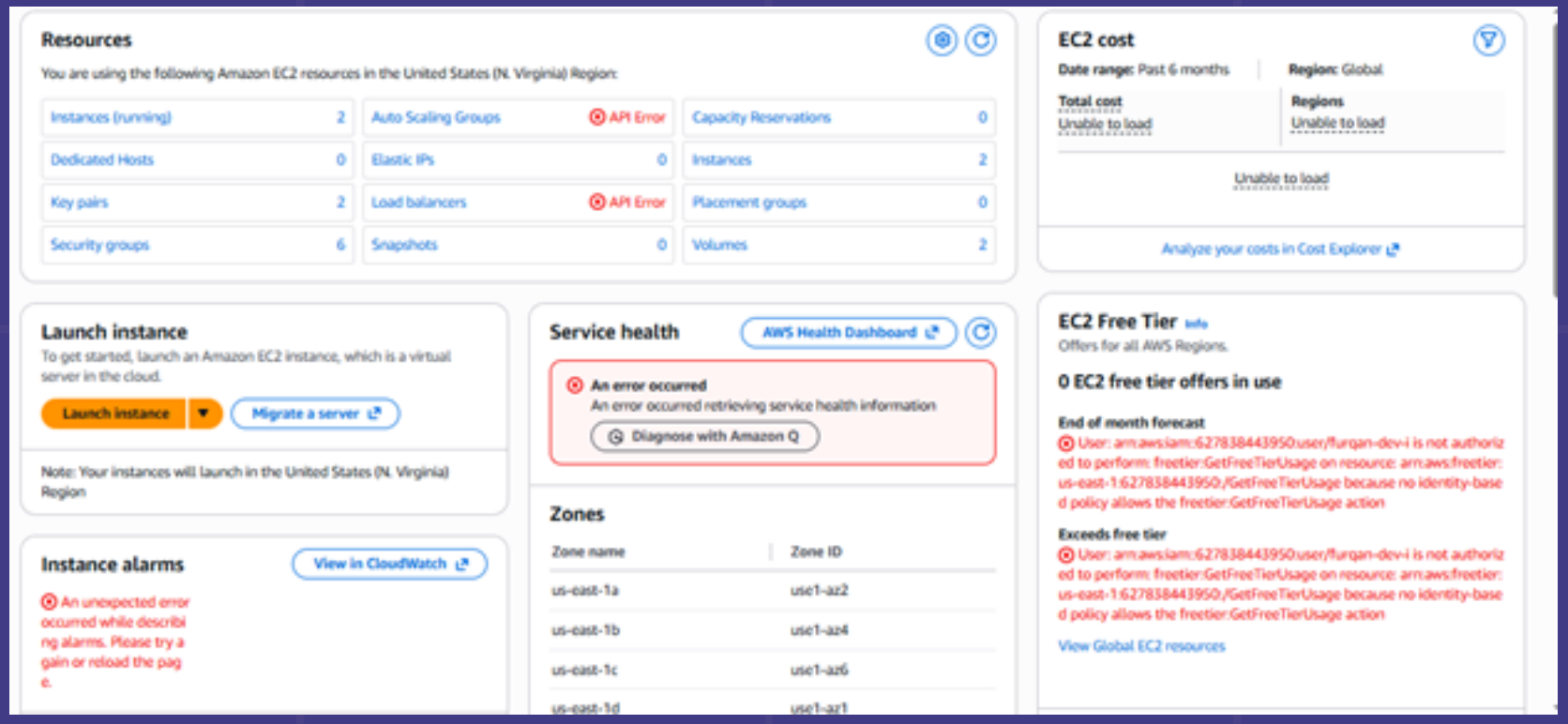
Testing the IAM Policy

Verifying Access Control

To test the policy, I logged in as the new IAM user.





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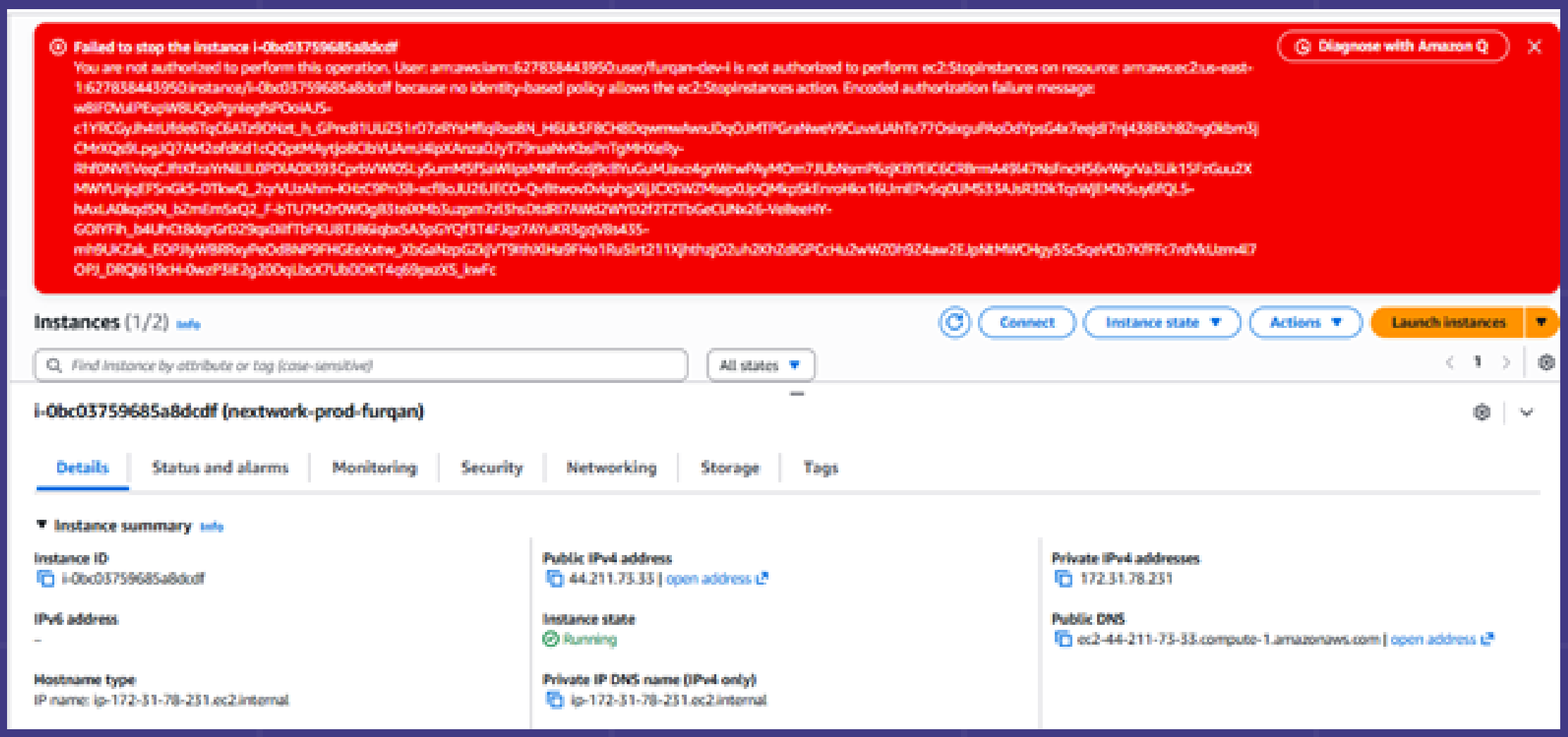
IAM Policies in Action

Testing Policy:

- Tried stopping the Production EC2 instance → Error (Not authorized)



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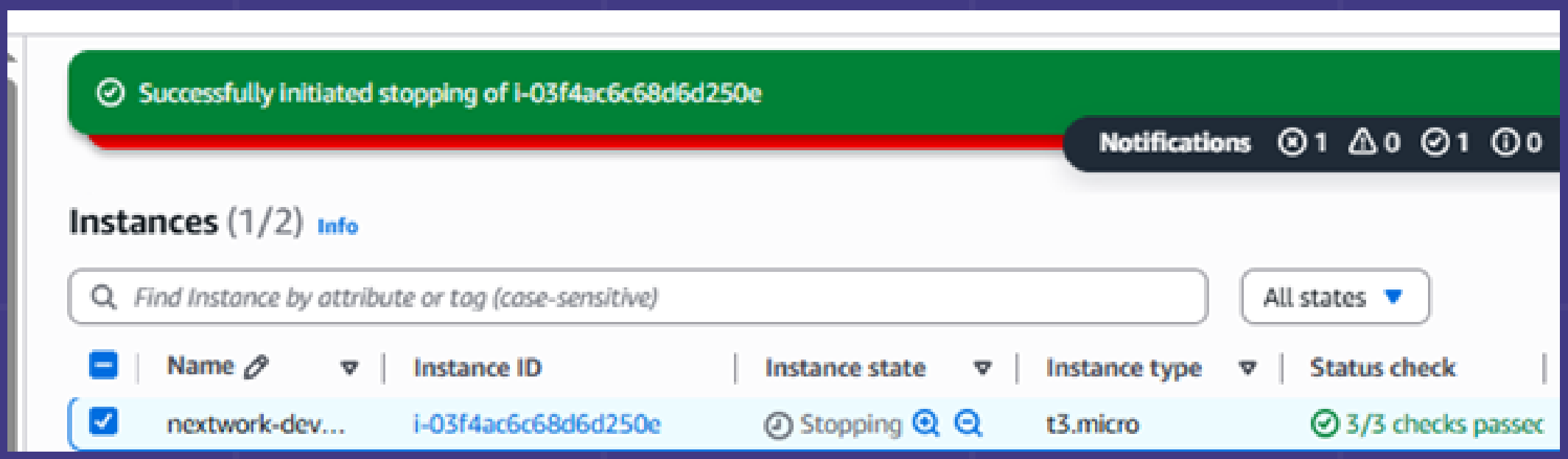
IAM Policies in Action

Testing Policy:

- Tried stopping the Development EC2 instance → Success (Allowed by policy)



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Key Learnings

- Launch and manage Amazon EC2 instances
- Use AWS IAM to enforce secure and precise access controls
- Automate permissions for new joiners using User Groups and Policies
- Customize AWS login with an Account Alias

Highlight: The project demonstrated how IAM policies work in real-world scenarios

📢 If you're into AWS or cloud security, let's connect!

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